CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Cache Organization

Lecture No.- 8



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Recap of Previous Lecture



Topic Block Replacement

Topic Miss Penalty

Topic Types of Cache Miss

Topics to be Covered



- **Topic** Hardware Implementation of Mappings
- **Topic** Hit Latency
- **Topic** Mux & Comparators in Cache
- **Topic** Array Access with Cache

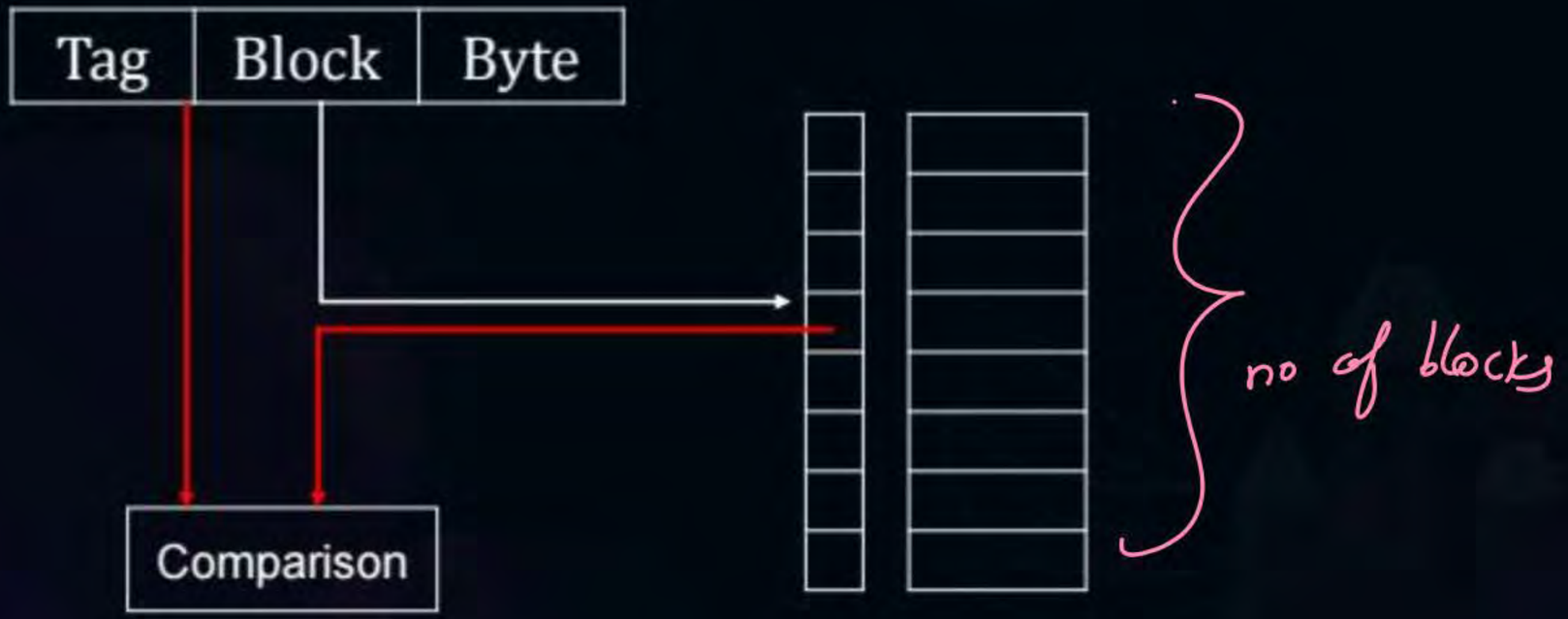


Topic : Checking Hit/Miss in Direct Mapping





Topic : Checking Hit/Miss in Direct Mapping





Topic : Hardware Implementation in Direct Mapping

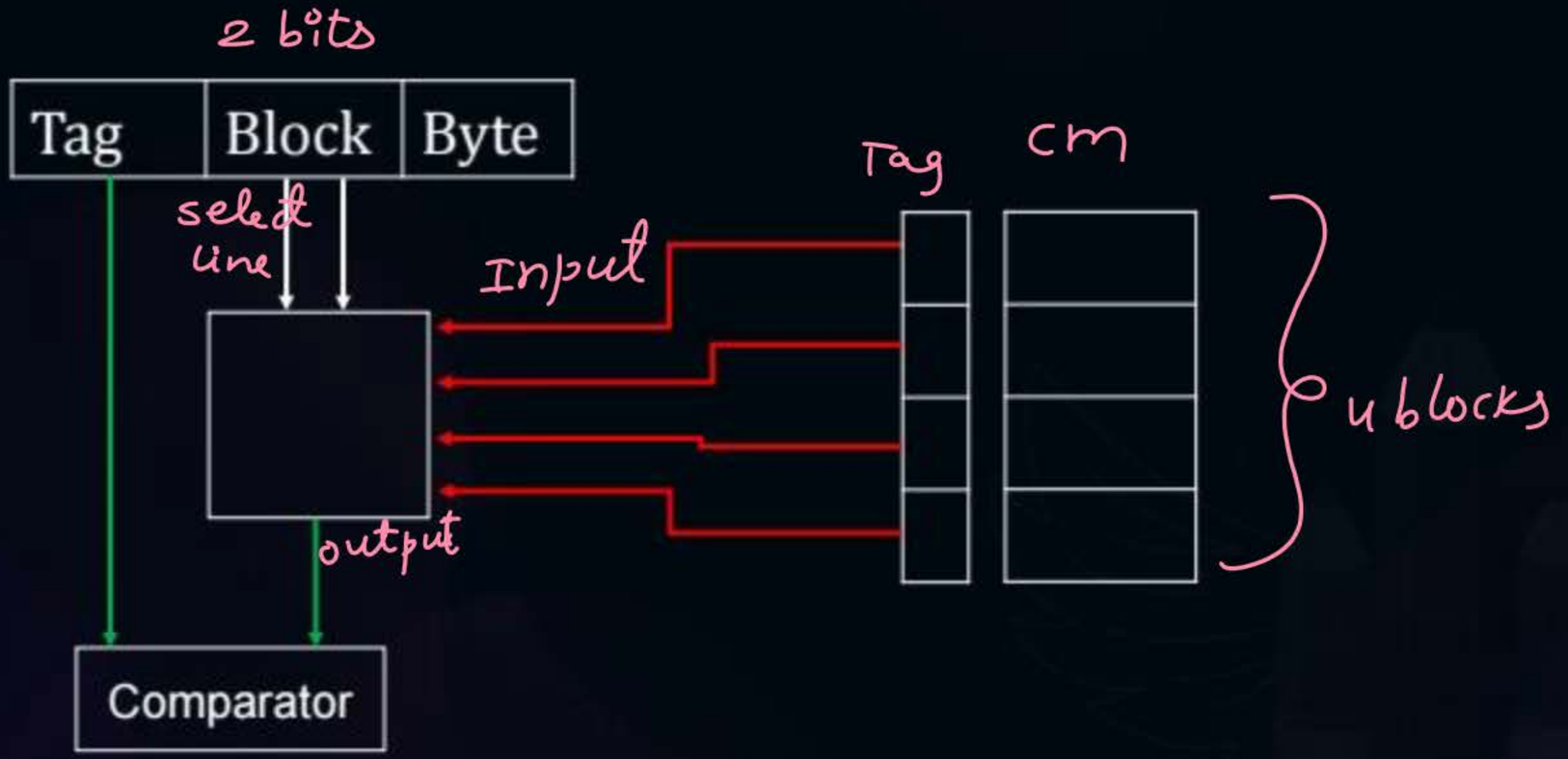
Blocks in cache = 4

Block size = 16 bytes

Main memory address = 8-bits



Topic : Hardware Implementation in Direct Mapping





Topic : Hardware Implementation in Direct Mapping





Topic : Hardware Implementation in Direct Mapping

No. of MUX needed for tag selection = No. of bits in tag

Size of each MUX — || — = No. of blocks : 1

Size of comparator = (no. of bits in tag) bit comparator

No. of — || — = 1



Topic : Hit Latency in Direct Mapping

Time taken by cache to decide it's a hit or miss

= MUX delay for tag selection + Comparator delay



Topic : Checking Hit/Miss in Set Associative Mapping

Tag	Set	Byte





Topic : Checking Hit/Miss in Set Associative Mapping



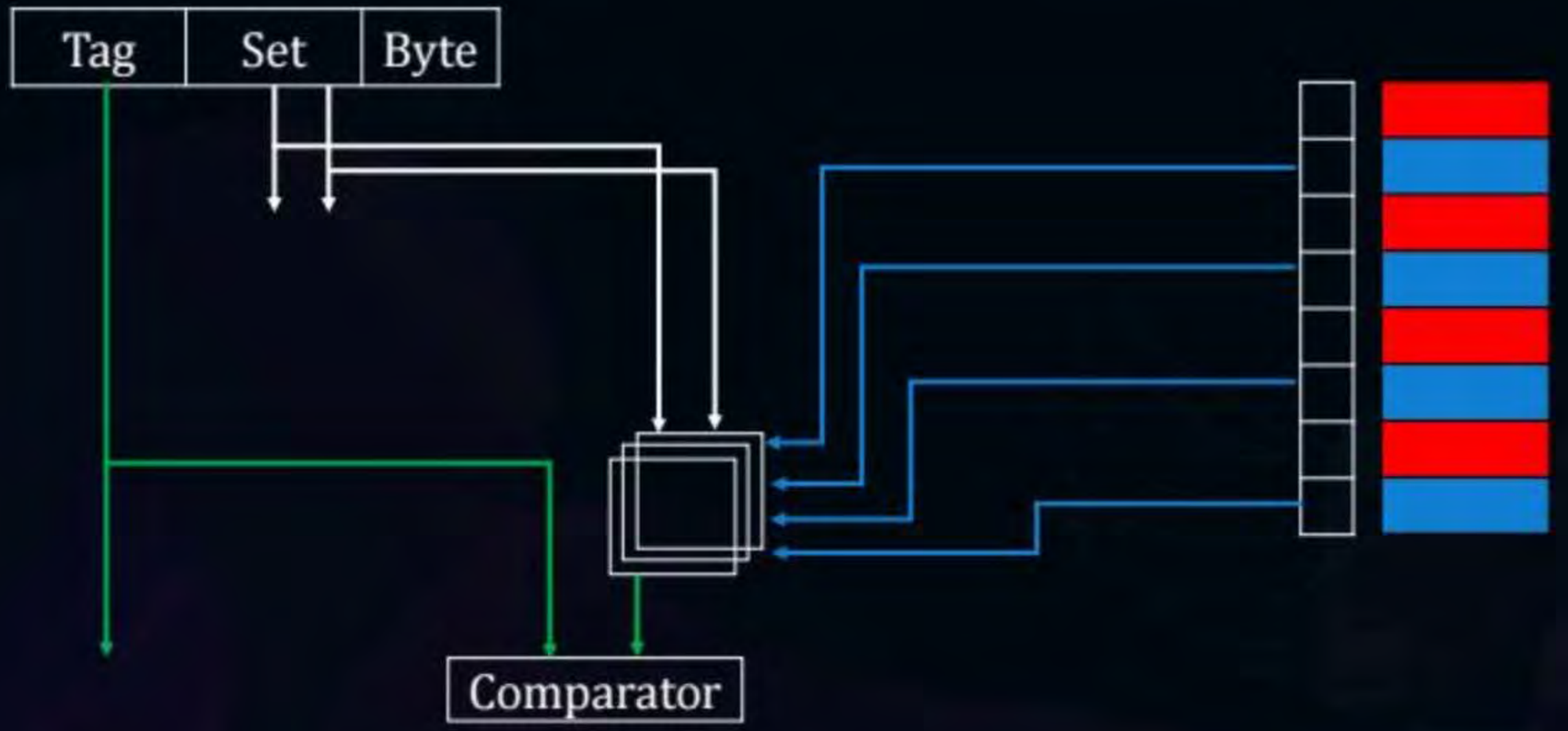


Topic : Hardware Implementation in Set Associative Mapping



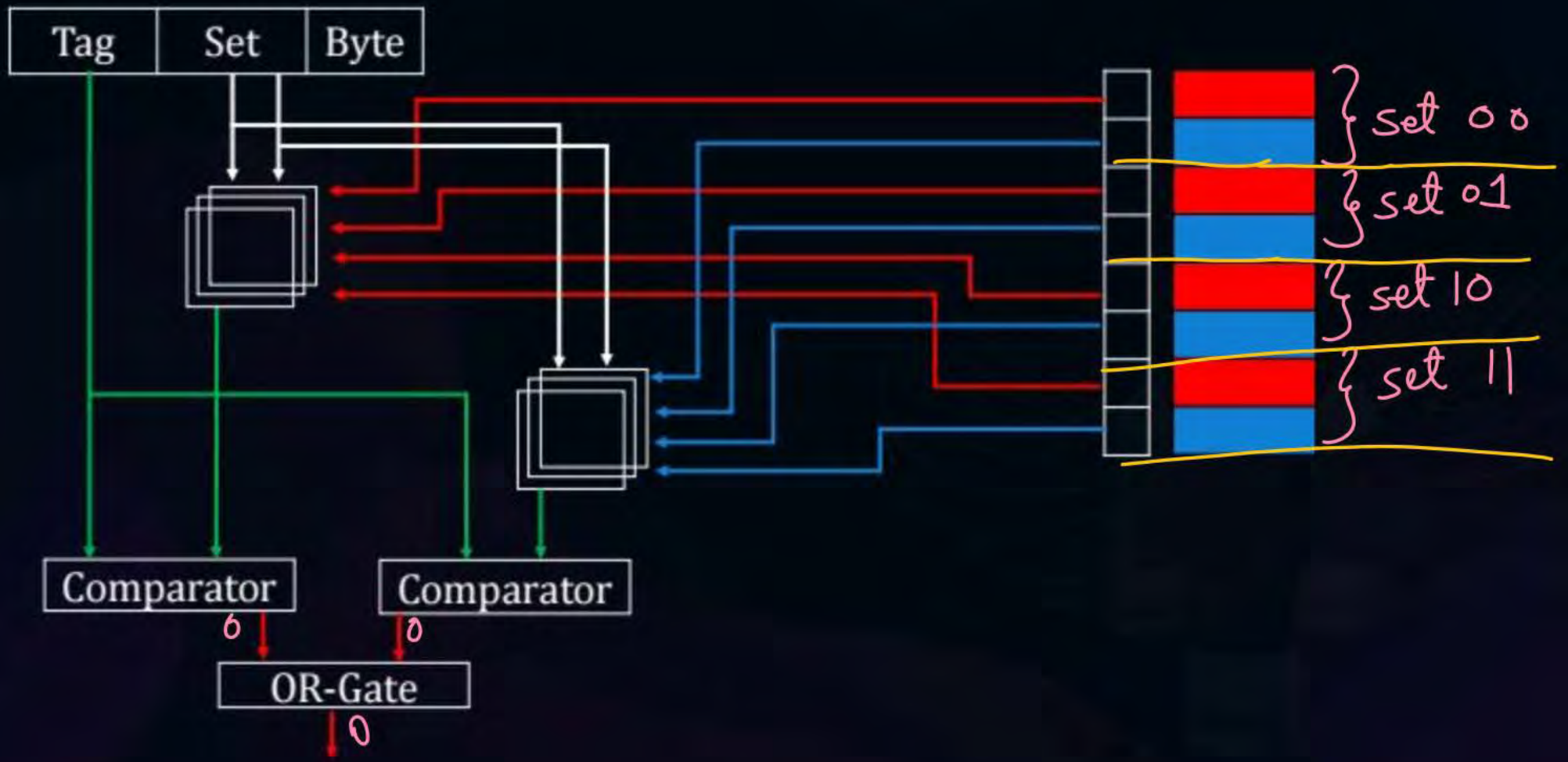


Topic : Hardware Implementation in Set Associative Mapping





Topic : Hardware Implementation in Set Associative Mapping





Topic : Hardware Implementation in Set Associative Mapping

No. of MUX for tag selection = k * no. of bits in tag

Size of each MUX ——— || ——— = no. of sets in cache : 1

No. of comparators = k

Size of comparators = (no. of bits in tag) bits comparator

No. of OR gates = 1

No. of inputs in OR gate = k

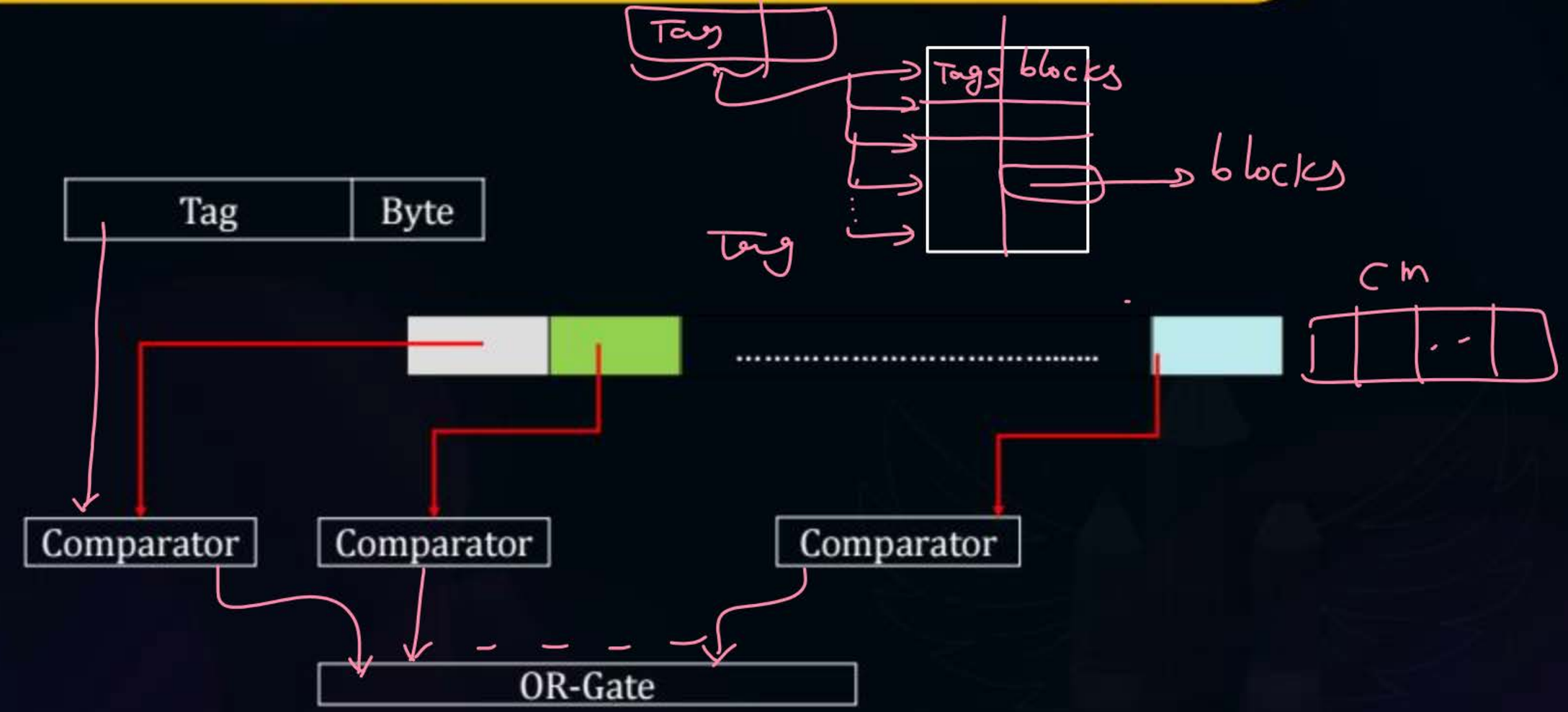
(k-way set ass. cache)



Topic : Hit Latency in Set Associative Mapping

= MUX delay for tag selection + Comparator delay + OR gate delay

Topic : Checking Hit/Miss in Fully Associative Mapping





Topic : Hit Latency in Fully Associative Mapping

Hardwares:-

no. of comparators = no. of blocks in cache

size of each comparator = (no. of bits in tag) bits comparator

OR gate ⇒ 1

no. of inputs ⇒ no. of blocks in cache

Hit latency = Comparator delay + OR gate delay

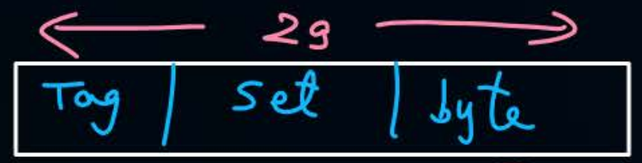
[NAT]

#Q. Cache Size = 128KB

Block size = 32 bytes $= 2^5$ B

Main memory address = 29-bits

4-way set associative cache



$\log 128k - \log 4$

$= 15$

no. of blocks in cache $= \frac{128k}{32k}$

$= \frac{2^{17}}{2^5} = 2^{12}$

1. Tag size? 14
2. Tag Directory size? $2^{12} * 14$ bits
3. Comparator required? 14 bits comparators, 4 needed
4. MUX required? $4 * 14 = 56$ MUX

   size $\Rightarrow 2^{10}:1$

[MCQ]

GATE-PYQ

#Q. Consider two cache organizations. First one is 32 KB 2-way set associative with 32-bytes block size, the second is of same size but direct mapped. The size of an address is 32 bits in both cases. A 2-to-1 multiplexer has latency of 0.6 ns while a k-bit comparator has latency of $\frac{k}{10}$ns. The hit latency of the set associative organization is h_1 while that of direct mapped is h_2.

The value of h_1 is:

2-way 32

Tag	set	byte
18	9	5

Direct 32

Tag	block	byte
17	10	5

log 32k = 15 bits

$h_1 = 0 + \frac{18}{10} + 0.6 = 2.4$ ns

A 2.4 ns ✓

B 2.3 ns

C 1.8 ns

D 1.7 ns

[MCQ] GATE - PYQ

#Q. Consider two cache organizations. First one is 32 KB 2-way set associative with 32-bytes block size, the second is of same size but direct mapped. The size of an address is 32 bits in both cases. A 2-to-1 multiplexer has latency of 0.6 ns while a k-bit comparator has latency of $\frac{k}{10}$ns. The hit latency of the set associative organization is h_1 while that of direct mapped is h_2.

The value of h_2 is:

A 2.4 ns

B 2.3 ns

C 1.8 ns

D ✓ 1.7 ns

$= 0 + \frac{17}{10}$

$= 1.7$ ns



Topic : Questions on Cache and Array

- Cache Size = 32 bytes
- Block size = 4 bytes

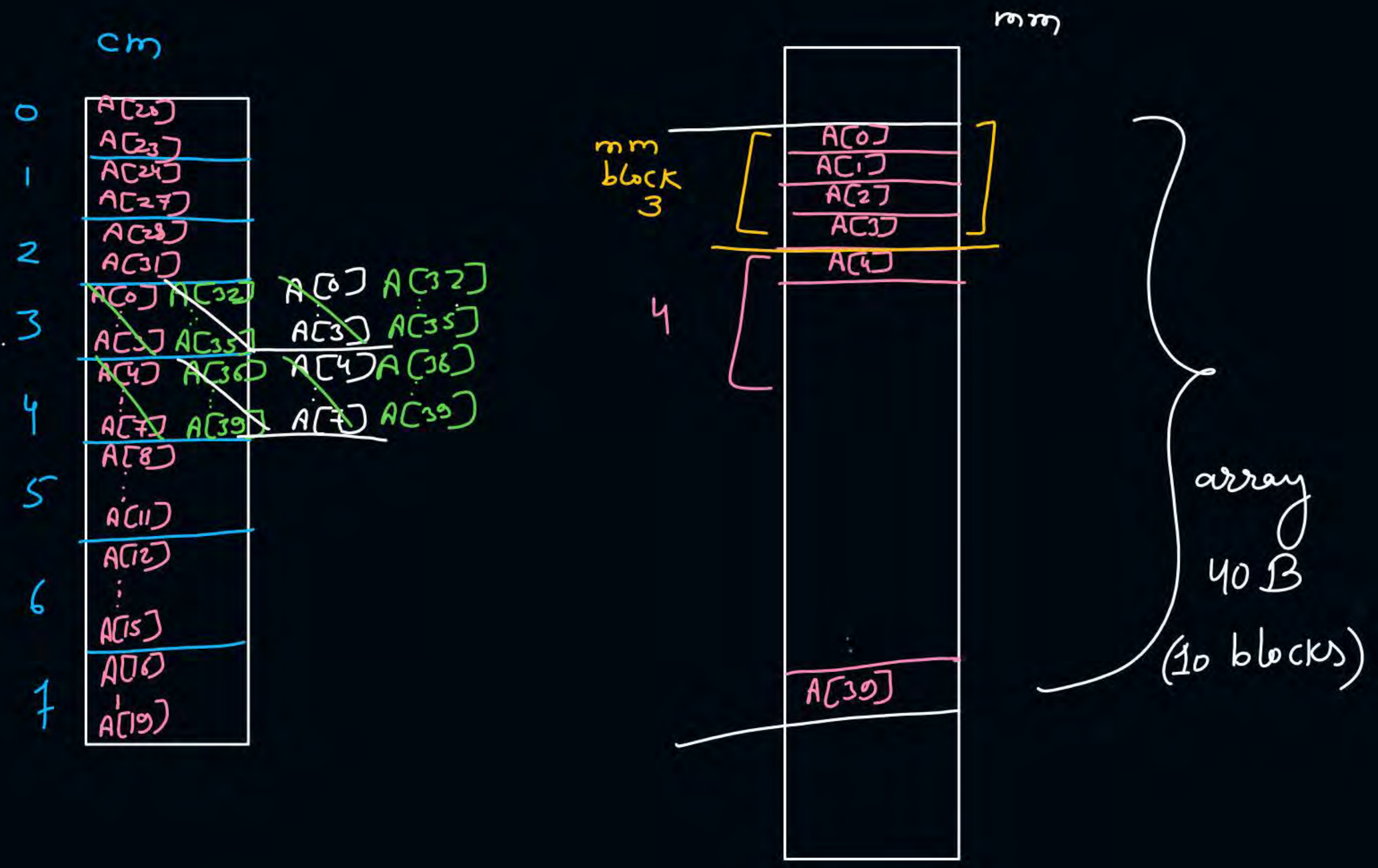
no. of blocks in CM = $\frac{32B}{4B} = 8$

- Array in main memory A[40], each element is 1 bytes
- Direct mapped cache

Size of array = $40 * 1 = 40B$

no. of blocks needed to store array = $\frac{40B}{4B} = 10$

No. of array elements per block = $\frac{4B}{1B} = 4$


cm
0
1
2
3
4
5
6
7
A[28]
A[29]
A[24]
A[27]
A[28]
A[31]
A[0] A[32]
A[3] A[35]
A[4] A[36]
A[7] A[39]
A[8]
A[11]
A[12]
A[15]
A[16]
A[19]
A[0] A[32]
A[3] A[35]
A[4] A[36]
A[7] A[39]
mm
mm block 3
4
A[0]
A[1]
A[2]
A[3]
A[4]
A[39]
array
40 B
(10 blocks)

CPU accesses	Hit/miss
A[0]	miss
A[1]	hit
A[2]	hit
A[3]	hit
A[4]	miss
A[5], A[6], A[7]	hit
A[8]	miss
A[9] ... A[11]	hit
A[12]	miss
A[13] ... A[15]	hit
A[16]	miss
A[17] ... A[19]	hit
A[20]	miss
A[21] ... A[23]	hit
A[24]	miss
A[25]	hit
A[27]	
A[28]	miss
A[29] .. A[31]	hit
A[32]	miss
A[33] .. A[35]	hit
A[36]	miss
A[37] ... A[39]	hit

2nd access	
A[0]	miss
A[1] .. A[3]	hit
A[4]	miss
A[5] .. A[7]	hit
A[8] ~ A[31]	hit
A[32]	miss
A[33] .. A[35]	hit
A[36]	miss
A[37] ~ A[39]	hit

	one time array access	2nd access	3rd access	4th access
no. of hits	30	36	36	36
no. of miss	10	4	4	4

no. of miss = no. of blocks in array

If array is accessed 4 times

Total hits =

miss =

no. of extra blocks in array = 10 − 8 = 2

	miss	hit
1st access	no. of blocks in array = 10	30
2nd access	2 * 2 = 4	36
3rd	2 * 2 = 4	36
4th	2 * 2 = 4	36
	22	138



Topic : Questions on Cache and Array

- Direct mapping
- Cache Size = 64 bytes
- Block size = 4 bytes
- Array in main memory int A[40], each element is 2 bytes

array size = 40 * 2 = 80 B

If array is accessed 4 times, no. of hits and miss ?

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no. of blocks in array = $\frac{80B}{4B} = 20$

no. of blocks in cache = $\frac{64B}{4B} = 16$

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no. of blocks extra in array = 20 − 16 = 4

	miss	hit
1st	20	20
2nd	4*2=8	32
3rd	8	32
4th	8	32
	44	116

$$\text{hit ratio} = \frac{116}{44+116}$$

$$= \frac{116}{160}$$

$$= 0.725$$

or

$$72.5\%$$

[NAT]

H.W.

#Q. Consider a machine with a byte addressable main memory of 2^{16} bytes. Assume that a direct mapped data cache consisting of 32 lines of 64 bytes each is used in the system. A 50 × 50 two-dimensional array of bytes is stored in the main memory starting from memory location 1100H. Assume that the data cache is initially empty. The complete array is accessed twice. Assume that the contents of the data cache do not change in between the two accesses.

How many data cache misses will occur in total?

[MCQ]

H.W.

#Q. Which of the following lines of the data cache will be replaced by new blocks in accessing the array for the second time?

A line 4 to line 11

B line 4 to line 12

C line 0 to line 7

D line 0 to line 8



2 mins Summary

Topic Hardware Implementation of Mappings

Topic Hit Latency

Topic Mux & Comparators in Cache

Topic Array Access with Cache



Happy Learning

THANK - YOU